



School of Engineering

Course : Diploma in Electronic & Computer Engineering (EGDF20)

Module : EGE322 – IoT System Project

Laboratory 6: Introduction to ThingSpeak

1. Objectives

This lab will introduce the ThingSpeak platform that allows users to send data to the cloud.

At the end of this laboratory session, student will be able to:

- Create Channel to view sensor data
- Use REST API to update status data

2. Material

Computer with Internet connection -1x

3. Theory

ThingSpeak is an open source Internet of Things (IoT) application and API to store and retrieve data from things using the HTTP protocol over the Internet or via a Local Area Network. ThingSpeak enables the creation of sensor logging applications, location tracking applications, and a social network of things with status updates that can be accessed from anywhere.

Students will create Channels to store all the data that a ThingSpeak application collects (temperature, humidity, luminosity and PIR). Each channel includes eight fields that can hold any type of data, plus 3 fields for location data and one for status data. Once data is collected, you can use ThingSpeak apps to analyse and visualize it. Sensor readings can be updated in the web browser using a URL request.

4. Procedures

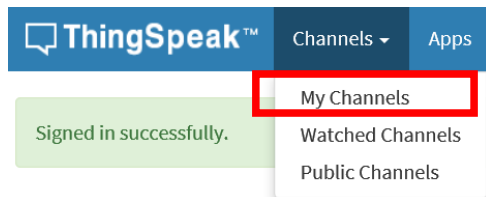
4.1 Register a MathWorks account in ThingSpeak

- Launch the Google Chrome browser and enter this url:
<https://thingspeak.com/>
- Click on Sign Up button to create MathWorks account in ThingSpeak.

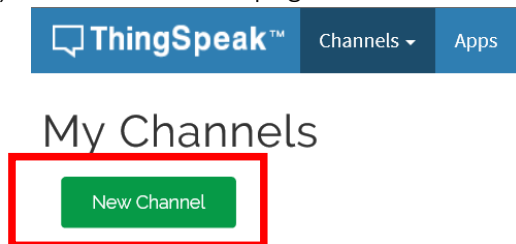
- Verify your MathWorks Account by clicking the link in the email ThingSpeak sent to you.

4.2 Create Channels and Dashboard

- Sign In to ThingSpeak with your MathWorks account at <https://thingspeak.com/>
- Click **Channels > MyChannels**.



- On the Channels page, click **New Channel**.



- Check the boxes next to Fields 1- 4. Enter these channel setting values:

Field	Label	Check
Field 1	Field Label 1	☒
Field 2		☐
Field 3		☐
Field 4		☐
Field 5		☐

Field	Label	Check
Field 1	Temperature	☒
Field 2	Humidity	☒
Field 3	Luminosity	☒
Field 4	PIR	☒
Field 5		☐

- Click **Save Channel** at the bottom of the settings.

f) You will now see the following tabs:

ThingSpeak™

Channels ▾

Apps

Community

Support ▾

Environmental Monitoring System

Channel ID: 359581

Sensor Data

Author: limkwaiv

Access: Private

Private View

Public View

Channel Settings

Sharing

API Keys

Data Import / Export

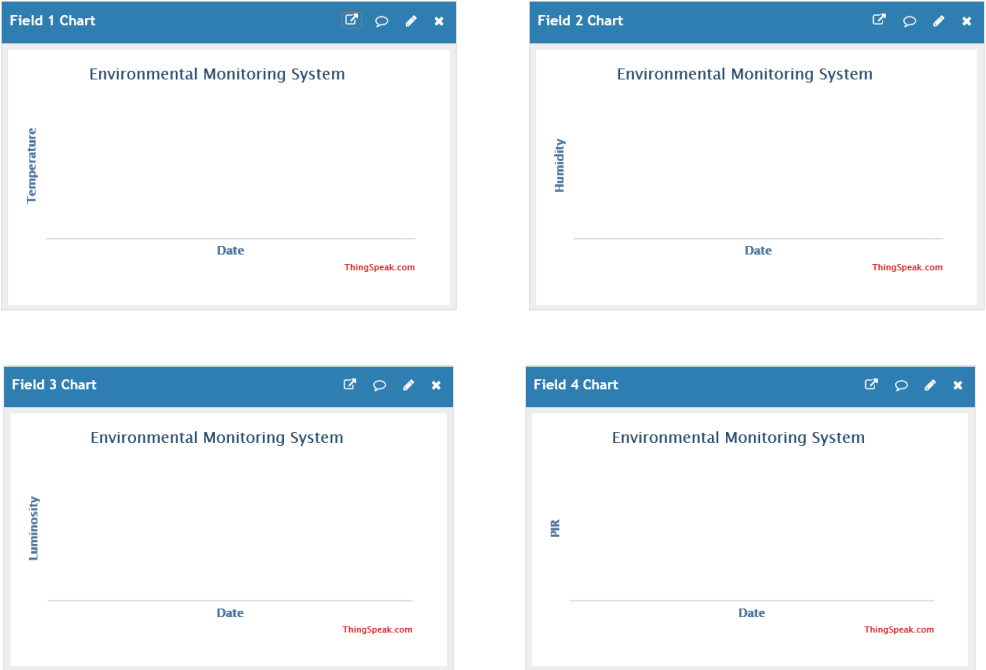
+ Add Visualizations

Data Export

g)

Private View	This tab displays information about your channel that only you can see.
Public View	If you choose to make your channel publicly available, use this tab to display selected fields and channel visualizations.
ChannelSettings	This tab shows all the channel options you set at creation. You can edit, clear, or delete the channel from this tab.
Sharing	This tab shows channel sharing options. You can set a channel as private, shared with everyone (public), or shared with specific users.
API Keys	This tab displays your channel API keys. Use the keys to read from and write to your channel.
Data Import/ Export	This tab enables you to import and export channel data.

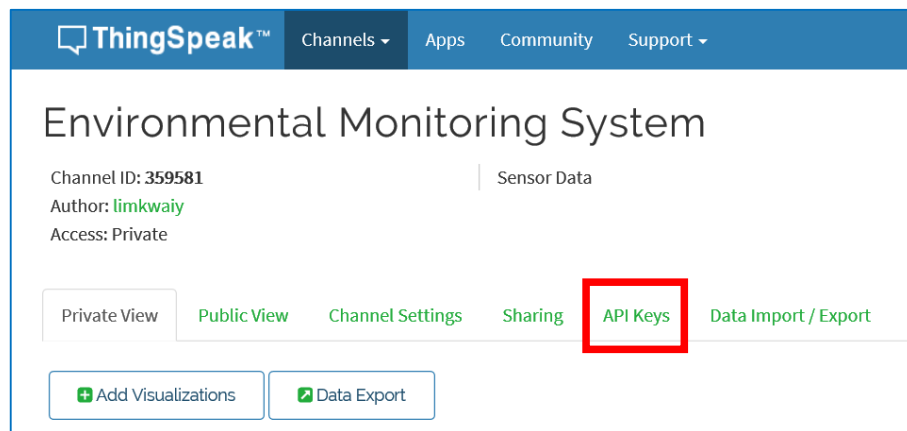
h) 4 Field Charts (Temperature, Humidity, Luminosity and PIR) are created as follows:



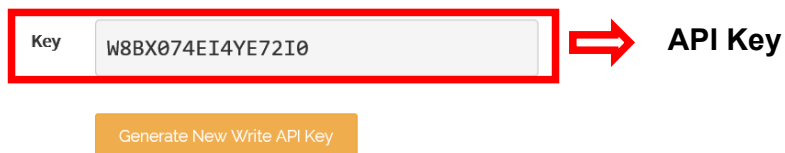
- i) Open another tab in your browser. Enter this URL to update the temperature in Field1:

`http://api.thingspeak.com/update?api_key=<api_key>&field1=xx`

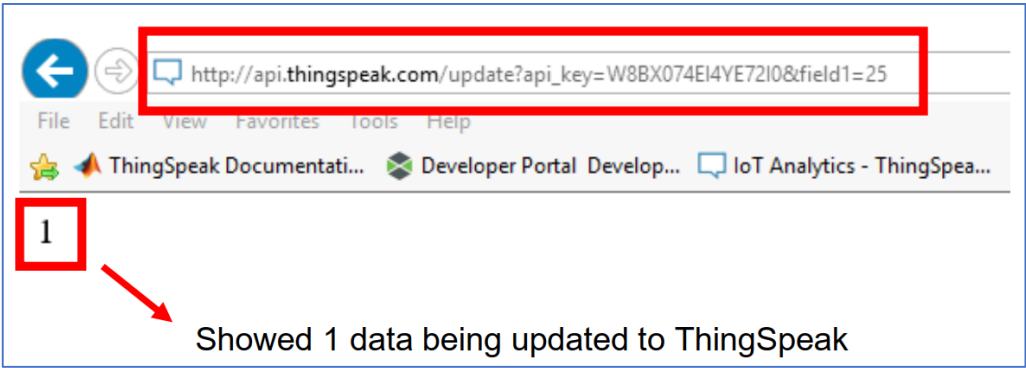
- Replace <api_Key> with your API Keys found in the tab of New Channel created in Step 2.6.
- To update value, replace value=xx with value=25.



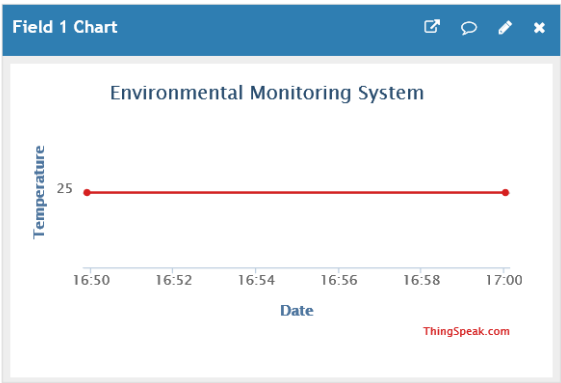
Write API Key



k) After the URL is entered, the browser will be updated as follows:



l) Refresh the browser tab that the Channel is in to reflect the change in Temperature.



m) You may repeat step 2.8, to update all the 4 fields (Temperature, Humidity, Luminosity and PIR) as follows:

